Submission Declarations — Electrical Engineering (Archiv für Elektrotechnik, Springer Nature)

Manuscript Information

Title: PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Structural Recovery for Resilient Power Transformer Thermal Dynamics Forecasting

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Availability of Data and Materials

The ETTh1 and ETTh2 benchmark datasets utilized in this study are standardized open-source time-series datasets from the Electricity Transformer Temperature (ETT) benchmark established by Zhou et al. (AAAI 2021). The datasets are freely accessible at: https://github.com/zhouhaoyi/ETDataset. No proprietary or restricted private data were used in this investigation.

Code Availability

Project Name: KAN-PSO-Transformer-Forecasting (PSO-TCWavKAN v9.1)

Repository URL: https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting

Complete source code, benchmark CSV results, ablation matrices, and statistical reproduction scripts are publicly accessible in the project repository at https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting under the MIT License. Detailed per-seed training logs and intermediate model checkpoints are archived in Kaggle Cloud environments and are available upon reasonable request from the author.

Operating System: Platform independent (verified on Windows 11 and Linux Ubuntu 22.04 LTS via Kaggle Cloud / NVIDIA Tesla T4 GPU)

Programming Language: Python 3.10 – 3.13

Primary Dependencies: PyTorch ≥ 2.0.0, NumPy ≥ 1.24, SciPy ≥ 1.10, Scikit-learn ≥ 1.3, Pandas ≥ 2.0, Matplotlib ≥ 3.7

License: MIT License (open source for academic and industrial research use)

Competing Interests

The author declares that there are no financial, personal, or professional competing interests that could have appeared to influence the work reported in this paper.

Funding

The author received no external funding, grant, or financial assistance from any public, commercial, or non-profit funding agency for this research. All computational resources were provided via Kaggle Cloud GPU environments.

Authors' Contributions

Nguyen Phu Toan is the sole author of this manuscript. The author conceived the mathematical formulation, implemented the architecture and optimization algorithms, designed and performed the empirical experiments, analyzed the benchmark results, and drafted and finalized the manuscript.

Ethics Approval and Consent to Participate

Ethics approval: Not applicable. This study does not involve human participants or animal subjects.
Consent to participate: Not applicable. Only publicly available open-source benchmark sensor datasets were used.

Consent for Publication

Not applicable. This manuscript contains no individual person's personal identifying data, images, or details in any form.